

Statistical Modelling (in the Social Sciences)
2022-2023
Practical Project 1

Please read carefully before you begin!

- This document is **3 pages** in total.
- In this assignment, you will be using data from the **Wave 20 of the 2014-2023 British Election Study Internet Panel**.
- The data, **in SPSS format**, and the questionnaire are available on Learn. You can also visit <https://www.britishelectionstudy.com> for other data types and more information.
- You are required to **use R** to carry out all the analysis! You can use any package or function which does the job. But for graphics, you **must use ggplot2**. Coding efficiency is important, but you won't be marked down unless it is terribly inefficient.
- Remember to check how **variables are coded** and whether you need any recoding or type conversion before your analysis. Treat all "Don't know"s, "Refused"s and such as missing. You are responsible to locate the variables using the Questionnaire if they are not explicitly stated in the Question.
- You should submit your answers as a **ZIP file** that contains your **Jupyter notebook file** or **R Markdown** document **and the knitted PDF**. If, for some reason, you cannot knit the document to PDF, you should submit a Word document + R Code. Your Word document should contain the screen shots of your codes and outputs together with your answers in text. In any case, briefly explain what you are doing in the code using **comments** and be clear about your variable choices – do not just give me variable names. You do **not need** to attach a **cover sheet**.
- Finally: **THIS IS NOT A GROUP PROJECT**. That means you are not allowed to discuss the assignment with anyone else. Please respect the rules of academic integrity.

- If you need any further clarification, please send me an email at ugur.ozdemir@ed.ac.uk

QUESTIONS

- Q.1) (Mark: 3/40)** What is the percentage of people who "strongly likes" either Boris Johnson or Nigel Farage? What is this percentage for males in Scotland? What is the percentage of who "strongly dislikes" Sturgeon and "strongly likes" Farage in England **Scotland**?
- Q.2) (Mark: 3/40)** What is the median likelihood of voting for Conservatives (**ptvCon**) among those who strongly dislikes Keir Starmer in Wales? What is the median probability of voting for Conservatives among those who strongly dislikes Starmer in England?
- Q.3) (Mark: 3/40)** What is the percentage of people who voted Labour in all the elections between 2010 and 2019? What percentage of these consistent **conservatives Labour voters** voted to stay in the EU?
- Q.4) (Mark:3/40)** Compare the distribution of the authoritarianism/libertarianism scores (variable: **al_scale**) for North-East, London and Scotland. Treat the variable as numeric/continuous. Briefly interpret this graph.
- Q.5) (Mark: 3/40)** On average, which country's response to coronavirus outbreak is found to be **least successful**? (Use variables benchCoronaXXX). What about those strongly likes the Conservative Party - which country do they find to be the most successful?
- Q.6) (Mark: 3/40)** What percentage of people aged between 20 and 30 ~~who~~ are very dissatisfied by the UK democracy? For which region this is maximum? What about those who are older than 60? **Answer these two questions for them as well.**
- Q.7) (Mark: 4/40)** Create a simple, additive measure of populist preferences using questions populism1 through populism6. Compare the descriptive statistics of this measure for the Conservative Party, SNP and Labour using tables and graphs. Interpret these findings.

- Q.8) (Mark: 4/40)** Are more populist people ~~happier~~ **more right wing**? Provide an answer to this question through a bivariate analysis using your measure from Question 7 and the **(lr_scale)** variable. Are they more ~~anxious~~ **authoritarian**? Run a similar analysis, this time using the **al_scale** variable.
- Q.9) (Mark: 6/40)** Choose the correct method to investigate the determinants of the left-right scores **(lr_scale)** treating it as a **numeric variable**. On top of age, gender, **and their interactions**, choose 3 more variables which you think make theoretical sense. (You will not be marked down for your variable choices unless they are totally irrelevant.) Comment on all the important output components of your model- coefficient estimates, significance levels, goodness of fit statistics etc. Run post-estimation diagnostic tests and interpret the results.
- Q.10) (Mark: 4/40)** Now consider your **model from Question 9**. What is the median prediction of your model for your sample? What about the median prediction of your model for the respondents from England? Is it higher or lower than the overall median? Compare using an appropriate graph. Is this compatible with the actual observations?
- Q.11) (Mark: 4/40)** Again, consider your **model from Question 9**. Compare the distribution of your residuals for London, Scotland and Wales using an appropriate graph. Are they different? What is your interpretation of the graph?